

Automatic Water Level Indicator and Pump Controller Using Discrete Electronic Components

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Abstract—Water wastage due to manual pump operation is a common problem in domestic water storage systems. Users often forget to switch off the water pump after the tank is full, leading to overflow and unnecessary loss of water. This paper presents a low-cost and reliable automatic water level indicator and pump controller using discrete electronic components without any microcontroller or Arduino. The system detects water levels at 40%, 70%, and 100% using conductive sensors and provides visual indication through LEDs. When the tank reaches 100% level, the pump is automatically turned off using a relay, and a buzzer alerts the user. The proposed system is simple, efficient, and suitable for household applications.

Index Terms—Water Level Indicator, Pump Automation, BC547 Transistor, 555 Timer IC, Relay Control.

I. INTRODUCTION

Water is one of the most essential natural resources, yet a large amount of water is wasted daily due to inefficient water management practices. In many households, water pumps are operated manually to fill overhead tanks. Due to human negligence, pumps are often left running even after the tank is full, causing overflow and wastage.

To overcome this problem, an automatic pump control system is required. Most modern solutions use microcontrollers, which increase cost and complexity. This project focuses on developing a fully hardware-based solution using basic electronic components. The proposed system automatically detects water levels and controls the pump without human intervention.

II. PROBLEM STATEMENT

The major issues addressed in this project are:

- Water wastage due to pump overflow
- Dependence on manual pump operation
- Lack of water level indication

The goal is to design an automatic system that is affordable, reliable, and easy to implement.

III. SYSTEM OVERVIEW

The proposed system consists of water level sensors, transistor switching circuits, LED indicators, a relay-controlled pump, and a buzzer alert system. Fig. 1 shows the system block diagram.

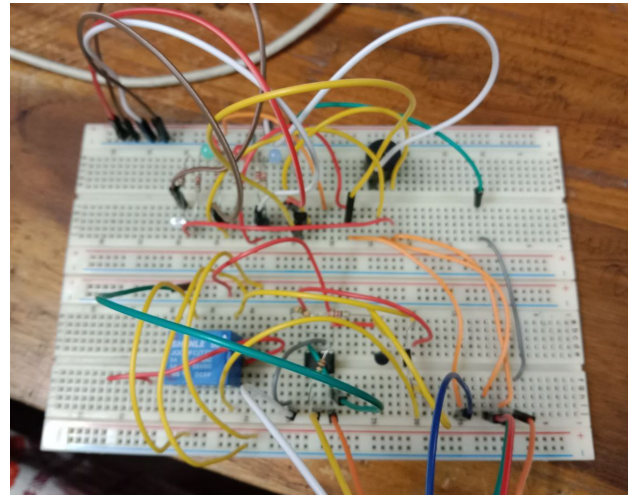


Fig. 1. The circuit connection of Automatic Water Level indicator and pump Controller

IV. HARDWARE COMPONENTS

A. Water Level Sensors

Conductive probes are placed at 40%, 70%, and 100% levels inside the tank. When water reaches a probe, it completes the circuit due to water conductivity.

B. BC547 Transistor

The BC547 NPN transistor is used as a switching device. Each water level probe controls one transistor, which activates LEDs, relay, or buzzer based on water level.

C. LED Indicators

Three LEDs indicate different water levels:

- LED1 – 40% water level
- LED2 – 70% water level
- LED3 – 100% water level

Each LED uses a 330Ω resistor for current limiting.

D. Relay and Water Pump

A 5V/6V relay is used to control the water pump. When the tank is full, the relay disconnects the pump automatically, preventing overflow.

E. 555 Timer IC and Buzzer

The 555 timer IC is configured to generate an audible signal through a buzzer when the tank reaches 100% level, alerting the user.

V. WORKING PRINCIPLE

The system operates based on simple transistor switching logic:

- 1) Initially, the pump is turned ON manually.
- 2) As water level rises, probes detect water contact.
- 3) At 40% and 70% levels, corresponding LEDs glow.
- 4) At 100% level:
 - The pump is automatically turned OFF using a relay.
 - The buzzer is activated.
 - All LEDs remain ON.

This ensures automatic control without user intervention.

VI. RESULTS AND PERFORMANCE ANALYSIS

The system was tested under different water levels. It responded accurately at all threshold levels. The pump was successfully turned off at full tank condition, eliminating overflow.

TABLE I
SYSTEM PERFORMANCE

Parameter	Observation
Level Detection Accuracy	High
Pump Response Time	Instant
Water Wastage	Eliminated
Power Consumption	Low

VII. ADVANTAGES

- No microcontroller required
- Low cost and simple design
- Automatic pump control
- Easy maintenance
- Suitable for domestic use

VIII. LIMITATIONS

- Works only with conductive liquids
- Fixed water level thresholds

IX. FUTURE SCOPE

Future improvements may include:

- Wireless monitoring
- Mobile notification system
- Solar-powered operation

X. CONCLUSION

This paper presented an automatic water level indicator and pump controller using discrete electronic components. The system effectively prevents water overflow, provides real-time level indication, and reduces human effort. Due to its simplicity and low cost, the proposed system is ideal for household water management applications.

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